

SpeakMate - Comprehensive Technical Analysis Report

1. Executive Summary

SpeakMate is an AI-powered English practice application designed specifically for Vietnamese students. The platform leverages a modern, distributed architecture combining a React/Next.js frontend, a Node.js/Express backend, and a high-performance Python-based AI pipeline hosted on Modal. It emphasizes real-time voice interaction (latency < 1s) through WebRTC (LiveKit) and complex scenario-based roleplay evaluated by specialized LLM agents.

2. System Architecture

SpeakMate utilizes a monorepo structure managed by npm workspaces. The system is divided into three primary execution planes:

- Frontend (Client Plane):** A Next.js 16 application running on Port 3000 handling UI, client-side WebRTC connections, user media capture, and optimistic state updates.
- Backend (Control Plane):** An Express 5 / Node.js application running on Port 3001 serving as the primary REST API, maintaining the single source of truth, handling business logic, orchestrating AI agents, and interfacing with the Supabase database.
- Cloud AI Pipeline (Data Plane):** Deployed on Modal Serverless infrastructure (GPU L4). It houses the heavy ML models: Speech-to-Text (STT) and Text-to-Speech (TTS), and connects directly to the LiveKit SFU (Selective Forwarding Unit) to stream audio frames to/from the user.

Core Pipelines

- WebRTC Real-time Pipeline (LiveKit):** The primary path (`NEXT_PUBLIC_USE_LIVEKIT=true`). Audio is streamed via User Mic -> WebRTC -> LiveKit Cloud -> Modal Python Worker. The worker performs VAD (Voice Activity Detection), STT (Faster-Whisper/PhoWhisper), calls Gemini for intent/text generation, and synthesizes audio via NeuTTS before streaming back to the client.
- HTTP Fallback Pipeline:** If LiveKit is disabled, audio is recorded on the browser via `MediaRecorder`, sent as `FormData` to the Express backend (Port 3001), which forwards it to the Modal HTTP endpoints or directly to Gemini API.

3. Technology Stack Breakdown

Frontend

- Framework:** Next.js 16 (App Router), React 19.
- Styling & UI:** Tailwind CSS v4, Radix UI primitives, shadcn/ui.
- Animations:** Framer Motion, tw-animate-css.
- RTC / Voice:** @livekit/components-react, @ricky0123/vad-web (Client-side VAD fallback).
- Forms & Validation:** react-hook-form, @hookform/resolvers (Zod).
- Charts:** Recharts (for radar charts in evaluations).

Backend

- Framework:** Express 5 web framework wrapped with TypeScript.
- LLM Interfacing:** @google/genai (Gemini SDK).
- Data Validation:** Zod ^3.25.76 (Strict middleware validation).
- Media Processing:** multer (in-memory storage), fluent-ffmpeg (.MP4 to .WAV transcoding for Safari support).
- RTC Control:** livekit-server-sdk (token generation and room grants).

Cloud AI Pipeline (Modal)

- Infrastructure:** Modal Serverless GPU functions.
- STT (Speech-to-Text):** Faster-Whisper / PhoWhisper (VinAI) using ctranslate2 for rapid inference.
- TTS (Text-to-Speech):** NeuTTS (dinhthuan/neutts-air-vi) via neucodec / bfloat16. (Previously Valtec ZeroShot TTS).
- RTC Worker:** livekit-agents and livekit-plugins.

Infrastructure, Data & Configuration

- Database & Auth:** Supabase (PostgreSQL, Supabase Auth via SSR, Supabase Storage for ephemeral 1-hour audio logs).
- Monorepo Tooling:** npm workspaces, tsx for typescript execution, cross-env.
- Shared Contracts:** Dedicated @speakmate/contracts package for synced TS interfaces.
- Local Dev Scripts:** PowerShell (service_wrapper.ps1, run-all.bat) with Named Mutexes and Signal files for persistent terminal management.

4. Multi-Agent Ecosystem

The business logic centers around domain-specific "Agents" instantiated strictly within the `PracticeController` or `StoryBankController`:

- BrainAgent:** Generates scenarios, evaluates contexts, adjusts goal difficulty, and creates gamified real-world challenges. (Returns JSON).
- VoiceAgent:** Handles intra-session conversational turns, interacting with the Modal pipeline or LLM Service. In Dual-character mode, decides which character speaks based on relevance.
- AnalystAgent:** Responsible for post-session grading. Grades across 9 sub-scores (vocabulary, empathy, persuasion, etc.) mapping to CEFR A1-C2 proficiency.
- MentorAgent (Mentor NI):** A chat interface providing targeted assistance in structuring "Story Bank" submissions based on the STAR/CAR/PREP frameworks.
- StoryBankAgent:** Processes raw chat logs and parses them into structured professional stories.

5. Project Directory Structure

```
d:\SpeakMate
|-- .ai/                                # AI assistant state logs and plans (SSoT)
|-- ai/                                # Python Cloud AI Pipeline (Deployed to Modal)
|   |-- agents/                        # Bridge logic between LLM and Audio Streams
|   |-- livekit_plugins/               # Custom STT/TTS wrappers
|   |-- modal/                        # Modal app definitions, webhooks, and workers
|-- backend/                           # Node.js / Express Control Plane
|   |-- src/
|       |-- agents/                   # Specialized LLM domain classes
|       |-- controllers/              # Route handlers
|       |-- middleware/               # Auth (JWT parsing) & Zod Body Validation
|       |-- routes/                   # API path definitions
|       |-- services/                 # Database, Storage, Audio, LiveKit, Auth logic
|       |-- validators/               # Request schemas (Zod)
|       |-- supabase/migrations/      # SQL DB schema updates
|-- frontend/                          # Next.js Application
|   |-- src/
|       |-- app/                      # Next.js App Router Pages
|       |-- components/               # Reusable UI/Shadcn/Practice Room components
|       |-- context/                  # React Context (Scenario, Language)
|       |-- hooks/                    # Custom hooks (useLiveKitRoom, useMentorChat)
|       |-- i18n/                     # Translations (EN/VI)
|       |-- lib/                      # Supabase SSR clients, apiClient
|-- packages/
|   |-- contracts/                    # Core TS interface definitions shared across Monorepo
|-- scripts/                           # Utilities for deployments and local developer setups
```